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Smart waste Collection & recycling management system

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Smart Waste Collection & Recycling Management System

Built on custom C++ data structures and graph algorithms — engineered to eliminate inefficient routing, overflowing bins, and skewed database lookups in municipal solid waste operations.

Core Problem & System Architecture

Real-World Challenges

- **Inefficient Routing:** High fuel waste from manual driver planning
- **Overflowing Bins:** Ultrasonic alerts skipped without fill-level prioritization
- **Skewed Databases:** Sequential indexing degenerates tree search to $O(N)$

System Workflow

01

Citizen Ticket
Submission

Custom FIFO
Queue — $O(1)$
enqueue

02

RFID / GPS
Asset Lookup

Chained Hash
Table collision
resolution

03

Bin Indexing
& Balancing

AVL Tree — $O(\log N)$ guaranteed

04

Truck Routing to Hot Zones

Dijkstra's Shortest Path on Graph

Core Data Structures & Code Implementation

Waste Priority Score & Heap Sort

Priority Score = waste Level $\times$ (capacity / 100.0) $\rightarrow$ current waste in Liters. Sorted in-place in **$O(N \log N)$** via custom Heap Sort.

Self-Balancing AVL Trees

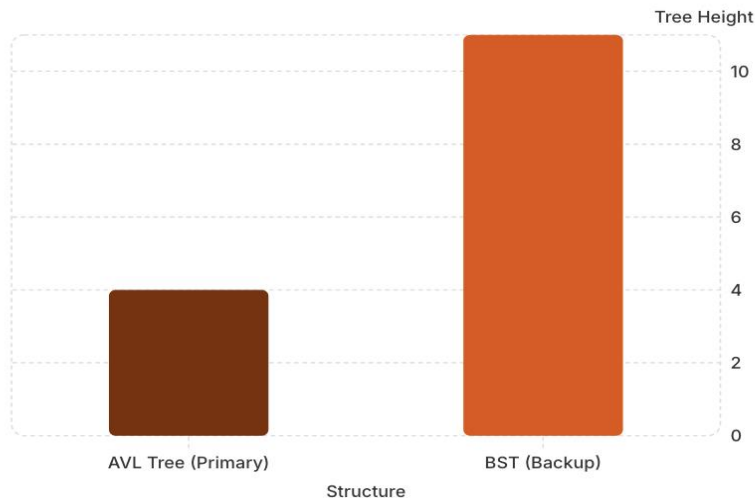
Balance Factor: $BF = \text{Height}(\text{Left}) - \text{Height}(\text{Right})$. If $|BF| > 1$, triggers rotations to maintain **$O(\log N)$** height.

Polynomial Rolling Hash Table

Hash Index = $\sum(\text{key}[i] \times 31^i) \% 53$. Collisions resolved via singly-linked chain buckets.

Performance Results & Technical Analysis

AVL vs. BST — Database Height Comparison



AVL maintains height 4 (guaranteed $\log N$). BST degrades to 11 from sequential inserts — a **63.6% reduction** in lookup iterations.

Recovery Stack & Environmental Report

LIFO Recovery Stack

Reverse-chronological action logging for rapid rollback

3,609 Liters

Total monitored waste volume

12.3% Recyclable

Share of recyclable waste captured

4 Urgent Bins

Flagged for immediate collection



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Thank You